Intergenerational Mobility and Credit

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These are our views and do not necessarily reflect the position of the Federal Reserve Bank of Minneapolis or the Census Bureau. Results were reviewed by the DRB.

Did Democratization of Credit Increase Intergenerational Mobility?

Democratization of credit. Cheaper bankruptcy & rising limits since 1970s

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   a. Expansion of credit limits: low earners increase child investment $\Rightarrow \uparrow$ mobility
   b. Lower bankruptcy costs: more pre-child borrowing, less investment $\Rightarrow \downarrow$ mobility

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   a. Expansion of credit limits: low earners increase child investment $\Rightarrow \uparrow$ mobility
   b. Lower bankruptcy costs: more pre-child borrowing, less investment $\Rightarrow \downarrow$ mobility (dominates)

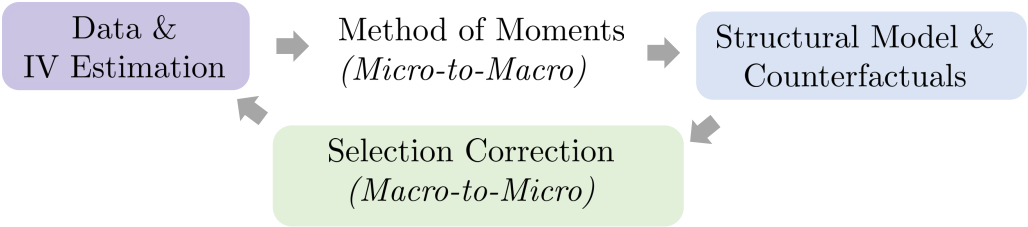
Answer. No. Credit democratization reduced mobility

Outline and Approach

Data & IV Estimation → Method of Moments *(Micro-to-Macro)* → Structural Model & Counterfactuals

1. Estimate causal effect of parental credit access
2. Discipline structural model using micro estimates

Outline and Approach



1. Estimate causal effect of parental credit access
2. Discipline structural model using micro estimates
3. Use model to correct IV selection
4. Simulate credit democratization

Data: Parental Credit Linked to Parents' & Child's Earnings

Census Data. Parent-child linkages (Decennial) + parent & child earnings (LEHD) +

TransUnion Credit Reports. Balance, limits, scores, bankruptcy status

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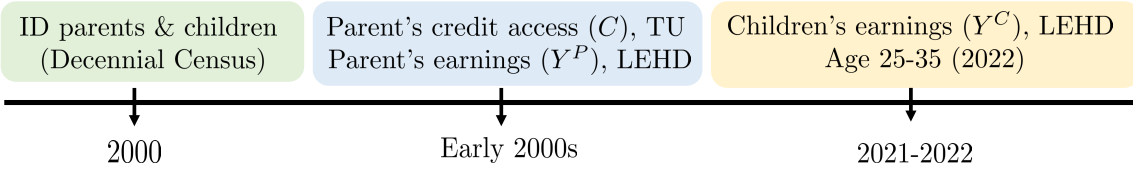
- Credit access (C) = unused revolving credit limits



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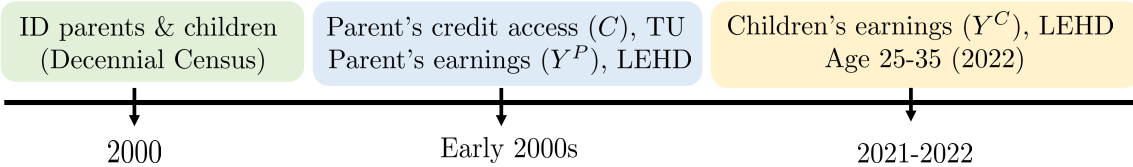


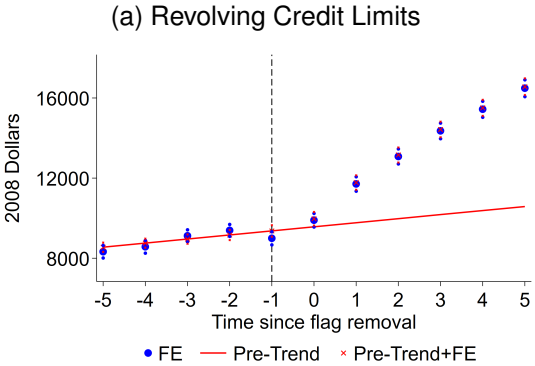
$$\log(Y_i^C) = \alpha + \beta \log(Y_i^P) + \eta \log(C_i) + \epsilon_i$$

β: Intergenerational Earnings Elasticity (IGE)

η: Intergenerational Credit Elasticity (ICE)

Bankruptcy Flag Removals

Flag removals. Follow statutory timelines (FCRA, 1970) & shift credit access



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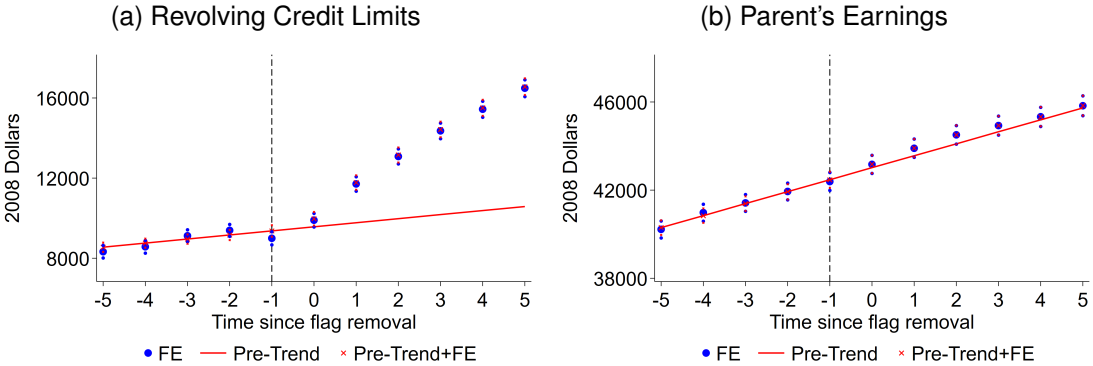
Flag removals. Follow statutory timelines (FCRA, 1970) & shift credit access, not earnings

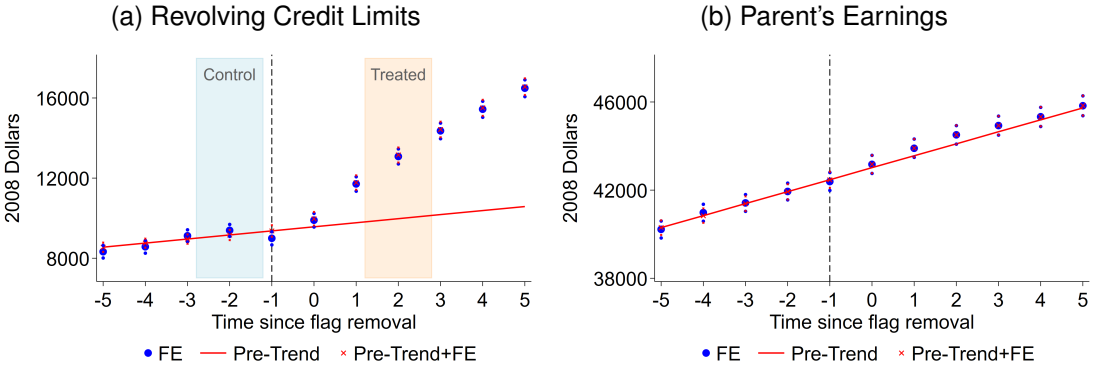


Bankruptcy Flag Removals

Flag removals. Follow statutory timelines (FCRA, 1970) & shift credit access, not earnings

Empirical approach ('05 cohort): 2003 removals (treated) vs 2007 removals (control)

- Instrument credit access in 2005 with flag removal indicator
- Repeat for 2003–2012 cohorts



Impact of Parental Credit Access on Children's Earnings

OLS Results. 10% increase in parental earnings $\Rightarrow$ 2.40% increase in child's earnings

	(1)
IGE	0.240***
	(0.00188)
ICE	
Observations	428,000
Controls	N
Sample	Full
Estimation	OLS

Impact of Parental Credit Access on Children's Earnings

OLS Results. 10% increase in parental earnings $\Rightarrow$ 2.40% increase in child's earnings

- $\uparrow$ parental credit access $\Rightarrow$ $\uparrow$ children's earnings

	(1)	(2)
IGE	0.240***	0.159***
	(0.00188)	(0.00203)
ICE		0.0338***
		(0.000328)
Observations	428,000	428,000
Controls	N	N
Sample	Full	Full
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Impact of Parental Credit Access on Children's Earnings

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- $\uparrow$ parental credit access $\Rightarrow$ $\uparrow$ children's earnings, controlling for wealth proxies

	(1)	(2)	(3)
IGE	0.240***	0.159***	0.0931***
	(0.00188)	(0.00203)	(0.00246)
ICE		0.0338***	0.0165***
		(0.000328)	(0.000346)
Observations	428,000	428,000	428,000
Controls	N	N	Y
Sample	Full	Full	Full
Estimation	OLS	OLS	OLS

Impact of Parental Credit Access on Children's Earnings

OLS Results. 10% increase in parental earnings ⇒ 2.40% increase in child's earnings

- ↑ parental credit access ⇒ ↑ children's earnings, controlling for wealth proxies

IV Results. 10% increase in parental unused credit ⇒ 0.64% increase in child's earnings

	(1)	(2)	(3)	(4)
IGE	0.240***	0.159***	0.0931***	0.0809***
	(0.00188)	(0.00203)	(0.00246)	(0.0117)
ICE		0.0338***	0.0165***	0.0641**
		(0.000328)	(0.000346)	(0.0230)
Observations	428,000	428,000	428,000	107,000
Controls	N	N	Y	Y
Sample	Full	Full	Full	BK Flag
Estimation	OLS	OLS	OLS	IV

Mechanisms

Greater credit access $\Longrightarrow$ more borrowing and better...

1. Investments in children
   - Higher childcare spending (TU-CPS ASEC)
2. Education outcomes
   - Higher college attendance & graduation
3. Labor market outcomes
   - Higher wages, higher paying firms, fewer non-employment spells
4. Ability to smooth large parental income declines
   - Lower pass-through of parental earnings shocks onto children's future earnings

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Interpretation. Greater parental credit access $\Longrightarrow$ higher child human capital

- Insulates child investment from parental shocks

Model Overview

Environment. Overlapping generations of dynastic, heterogeneous households

- Parents differ in human capital h, assets b, & bankruptcy flag $k \in \{C, N\}$
- Children's initial human capital h^c depends on parental human capital

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Choices. Parents choose child investment i, save/borrow, default

- Default $\Rightarrow$ utility penalty $\psi(b)$ & bankruptcy flag ($k = N$)

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Shocks. Income (and expense shocks)

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Shocks. Income (and expense shocks)

Credit Market. Small open economy, individually priced debt

+ Exogenous limit $\underline{b}_k(w(h))$, tighter for bankrupt

Parental Investment Stage

Parents w/ credit access (age 28-47): invest i in child human capital & save/borrow

$$V_j^C(b,h,h^c) = \max_{b',i\geq 0} \; u(c) + \beta\mathbb{E}\left[\max\{\underbrace{V_{j+1}^C(b',h',h^{c'})}_{\text{Repay}}, \underbrace{V_{j+1}^N(0,h',h^{c'}) - \psi(b')}_{\text{Default}}\}\right]$$

Constraints.

$$c + \underbrace{q_{j,C}(b',h,h^{c'})}_{\text{bond price}} b' + i \leq w(h) + b; \qquad b' \geq \underbrace{\underline{b}_C(w(h))}_{\text{limit}}$$

Human Capital Laws of Motion.

$$h' = h + \eta, \;\; \eta \sim N(\mu_\eta, \sigma_\eta^2); \qquad h^{c'} = (1-\omega_c)h^c + \omega_c \log\left(\frac{i+d}{\zeta_c}\right)$$

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Child leaves (parent age 48-51, child age 20-23): altruism θ governs one-time transfer τ

Micro-to-Macro: Calibration

External. 4-year model period, CRRA utility $\sigma = 2$, annual risk-free rate 4%

Human capital. $h^c = \rho_c h + \eta_c \qquad h^{c'} = (1 - \omega_c) h^c + \omega_c \log\left(\frac{i+d}{\zeta_c}\right)$

Variable	Value	Target	Model	Data	Source
A. Human Capital					
ρ_c	0.150	Intergenerational earnings elasticity (IGE)	0.258	0.240	TU-LEHD-Dec
ω_c	0.090	Intergenerational credit elasticity (ICE)	0.067	0.064	TU-LEHD-Dec
ζ_c	0.529	Investment to earnings, age 0-3	0.022	0.033	Lee & Seshadri (2019)

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Credit Market. $\psi(b) = \psi \cdot b \qquad \underline{b}_C(w(h)) = \alpha_C + \delta_C \times w(h)$

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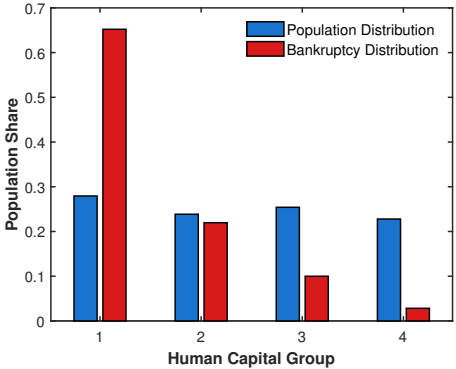
B. Credit					
ψ	8.445	Bankruptcy rate	3.146	3.319	ABI 2001-2004
α_C	-0.086	Average credit limits to earnings	0.255	0.255	SCF 2001-2004
δ_C	-0.204	Slope, change in limits to change in earnings	0.204	0.204	SCF 2001-2004
β (Annual)	0.901	Agg. credit to earnings	0.025	0.026	SCF 2001-2004

Macro-to-Micro: IV Selection

Concern. Bankruptcy concentrated among low human capital parents



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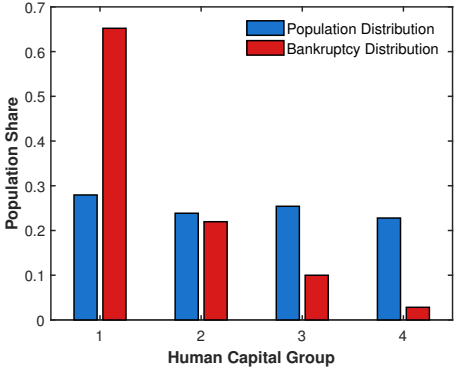
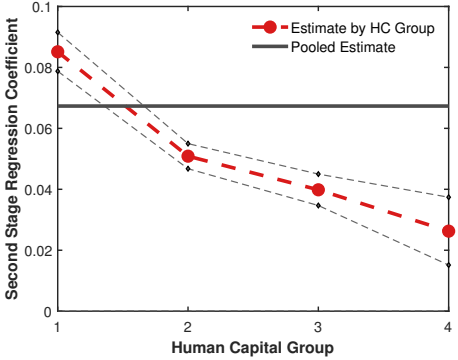
Correction. Estimate IV by human capital quartile

- Largest effect for lowest human capital



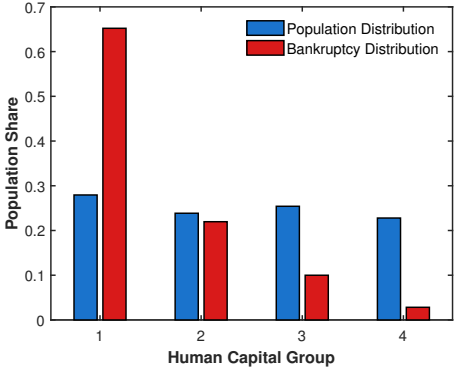


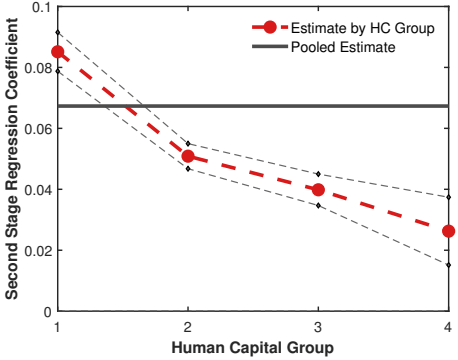
Macro-to-Micro: IV Selection

Concern. Bankruptcy concentrated among low human capital parents

Correction. Estimate IV by human capital quartile

- Largest effect for lowest human capital
- Re-weight by population shares
- Baseline IV is 20% too large





Experiment: Democratization of Credit from 1970s to 2000s

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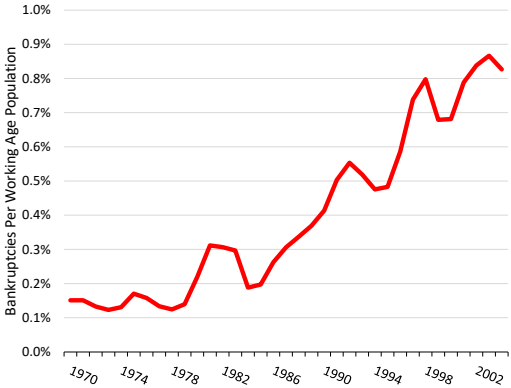

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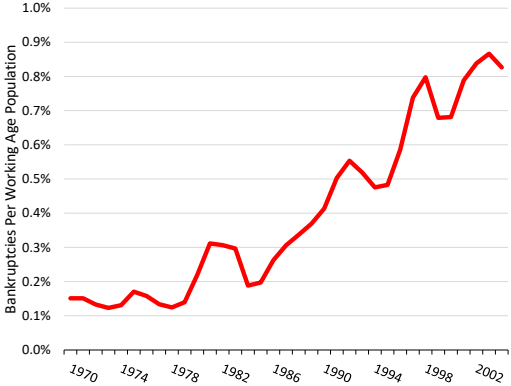
1. BK penalty higher to match $6x$ lower BK rate from ABI: $\psi_{70} \approx 6\psi_{00}$
2. Limits tighter to match SCF: $\alpha_{C,70} \approx 0$, $\delta_{C,70} \approx \frac{1}{4}\delta_{C,00}$



Counterfactual: Credit Democratization

Results:

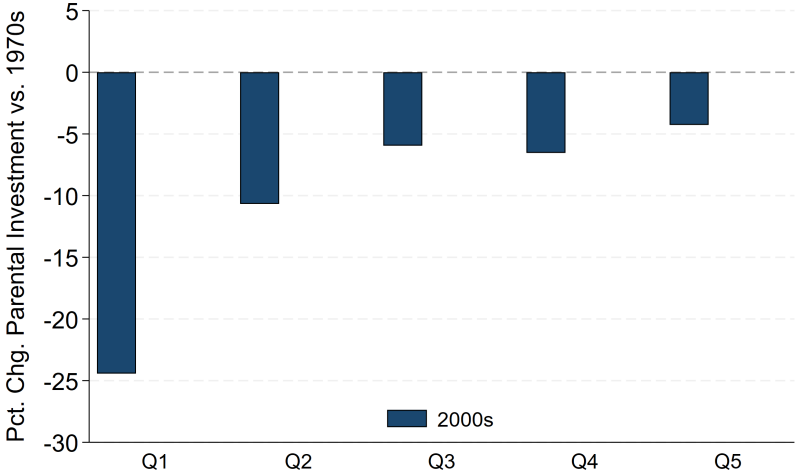
i. Reduced intergenerational mobility

ii. Increased earnings and consumption inequality

	1970s Economy	2000s Economy	% Change
Intergenerational earnings elasticity (IGE)	0.238	0.258	+8.5%
Variance log earnings, age 24–27	0.392	0.402	+2.6%
Variance log consumption	1.273	1.341	+5.3%

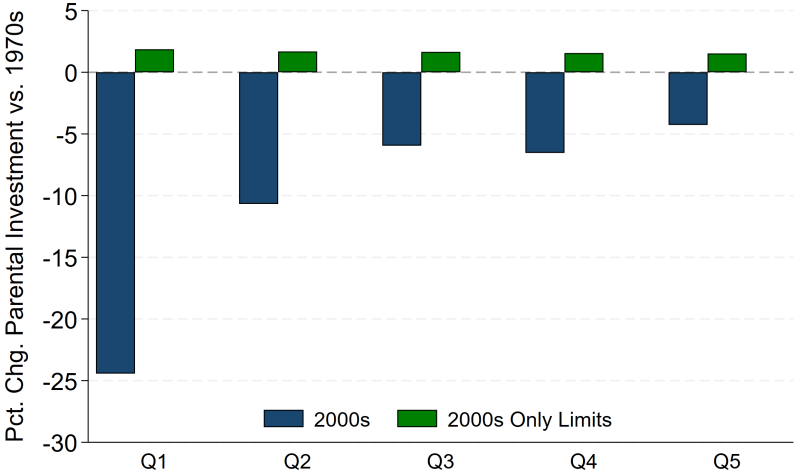
Why Has Democratization of Credit Markets Reduced Mobility?

Parental investment decreases among low earners



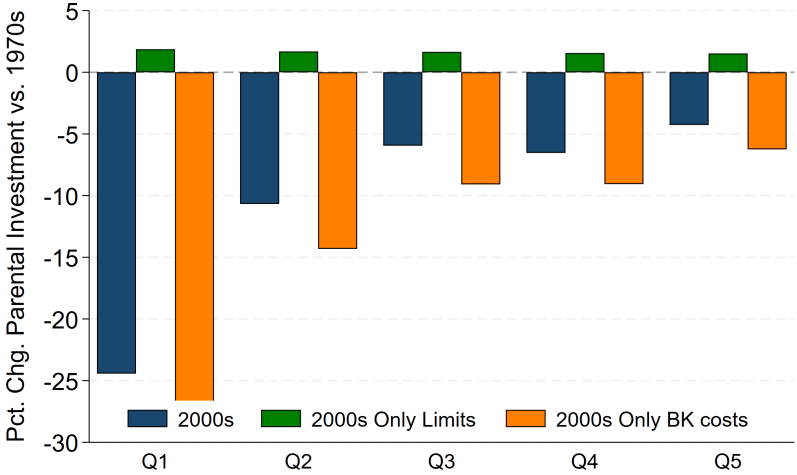
Why Has Democratization of Credit Markets Reduced Mobility?

Parental investment decreases among low earners, not due to ↑ limits



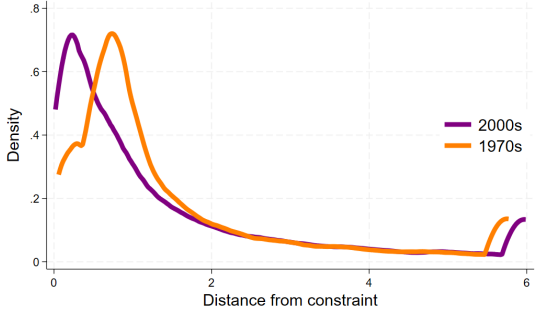
Why Has Democratization of Credit Markets Reduced Mobility?

Parental investment decreases among low earners, due to ↓ bankruptcy costs



Why Has Democratization of Credit Markets Reduced Mobility

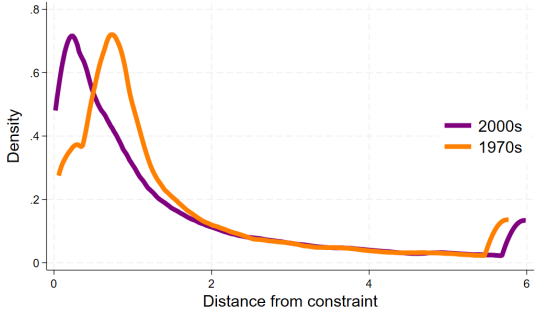
i. Households move closer to their borrowing constraints

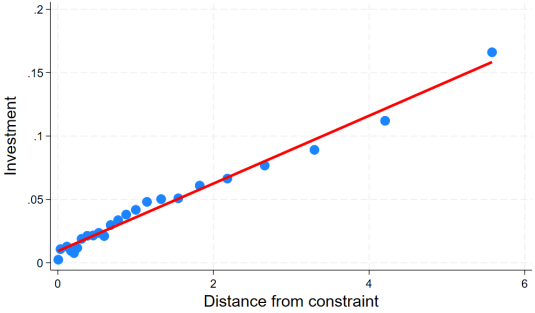


Why Has Democratization of Credit Markets Reduced Mobility

i. Households move closer to their borrowing constraints

ii. Child investment declines as households approach the constraint





Conclusion

Contributions:

i. Greater parental credit access increases children's earnings

ii. Develop a dynastic model with defaultable debt

iii. Credit democratization since the 1970s has reduced mobility by lowering child investment among low-income households

Additional Results

- Age of oldest account instrument.
- Placebo test flag removal.
- Flag removal by age.
- Childcare Results.
- Education & Labor Market Results.
- Smoothing Results.
- Model Validation.
- Credit Experiment Validation.
- IGE by Bankruptcy Costs.

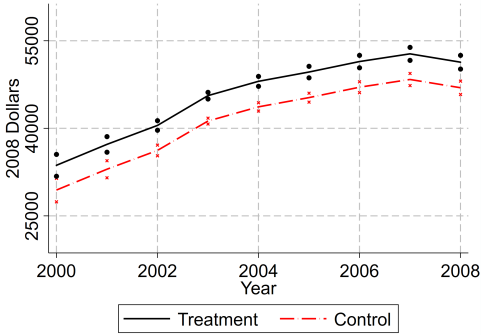
Additional Instrument: Age of oldest credit account

Age of oldest credit account has persistent impact on credit access but not earnings

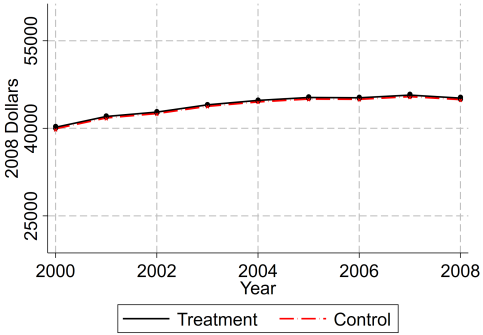
Empirical approach: Compare HH w/ first credit card in 1970 (treatment) vs. 1972 (control)

- Instrument parent's credit access in 2005 with treatment indicator
- Repeat for 1971/1973, 1972/1974,..., 2000/2002 comparisons ....

(a) Revolving Credit Limits



(b) Earnings



Back to Summary

Age of Oldest Account: Results

Result: 10% increase in parental unused credit, 0.5% increase in child's earnings

	(1)	(2)	(3)
	Dependent variable: log of child's earnings		
Log Parent's Earnings	0.0539***	0.0524***	0.0393***
	(0.00666)	(0.00674)	(0.00889)
Log Unused Revolving Credit	0.0545***		
	(0.00665)		
Log Revolving Credit Limit		0.0532***	
		(0.00676)	
Credit Score			0.494***
			(0.0630)
Observations	855000	855000	855000
Baseline, Wealth, Type Controls	Y	Y	Y
Interest & Dividend Controls	N	N	N
Sample	Main	Main	Main
IV Strategy	Stacked	Stacked	Stacked

Back to Summary

Age of Oldest Account: Results

Result: 10% increase in parental unused credit, 0.5% increase in child's earnings

- Robust to controls for interest & dividend income

	(1)	(2)	(3)	(4)	(5)
	Dependent variable: log of child's earnings				
Log Parent's Earnings	0.0539***	0.0524***	0.0393***	0.0564***	0.0595***
	(0.00666)	(0.00674)	(0.00889)	(0.00874)	(0.00799)
Log Unused Revolving Credit	0.0545***			0.0475***	0.0439***
	(0.00665)			(0.0104)	(0.00528)
Log Revolving Credit Limit		0.0532***			
		(0.00676)			
Credit Score			0.494***		
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Observations	855000	855000	855000	133000	67000
Baseline, Wealth, Type Controls	Y	Y	Y	Y	Y
Interest & Dividend Controls	N	N	N	Y	Y
Sample	Main	Main	Main	Main	Main
IV Strategy	Stacked	Stacked	Stacked	Stacked	Non-Stacked

Back to Summary

Flag Removal: Exposure and Placebo

- Compare exposure to good credit before age 18 vs. after age 18
- Let $g_i = 18 - a_{i,remove}$ denote child i's good exposure before 18
- Positive g_{it} means flag removal before the child turned 18. Negative means after 18.

$$\log(Y_i) = \beta \log(Y_i^P) + \eta^{(+)} \cdot g_i \cdot 1(g_i > 0) + \eta^{(-)} \cdot g_i \cdot 1(g_i \leq 0) + \Gamma X_i + \epsilon_i$$

	(1)	(2)	(3)	(4)
	Dependent variable: Log of child's earnings			
Parents earnings	0.202***	0.114***	0.113***	0.113***
	(0.00518)	(0.00590)	(0.00590)	(0.00590)
Years of good credit before 18, $\eta^{(+)}$			0.00197**	0.00206**
			(0.000876)	(0.000890)
Years of good credit after 18, $\eta^{(-)}$			0.00320	0.0103
			(0.00503)	(0.0130)
Indicator flag removed after 18				0.0124
				(0.0209)
R-squared	0.019	0.206	0.206	0.206
No. Obs	84000	84000	84000	84000
Sample	DF	DF	DF	DF
Baseline Control	N	Y	Y	Y
Wealth Control	N	Y	Y	Y

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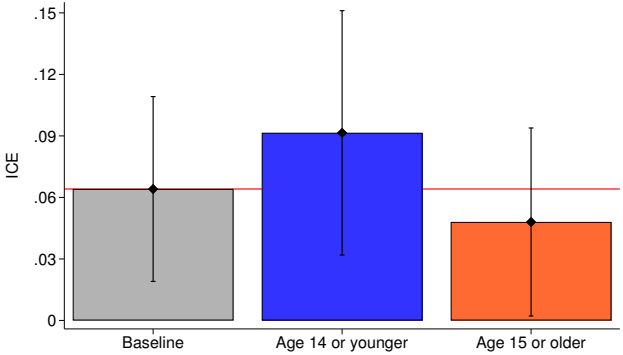
Flag Removal: Heterogeneity by child age

- Stratify by age of child ($<$14yo) at flag removal – larger effects when younger
- However p-value of difference is 0.09



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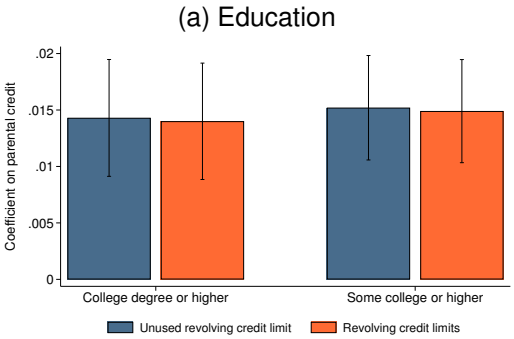
Education

- Regress college attendance indicator on credit access (AOA IV)
- More likely to attend college/some college (10% more credit, 0.15pp increase in college)



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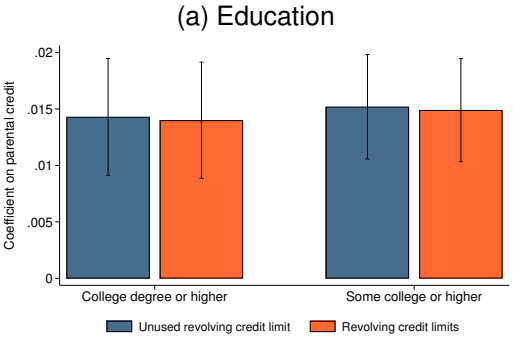
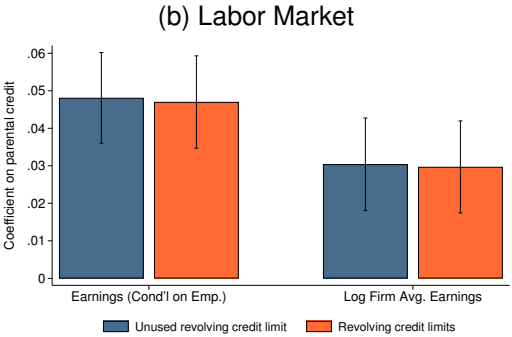
Education

- Regress college attendance indicator on credit access (AOA IV)
- More likely to attend college/some college (10% more credit, 0.15pp increase in college)
- Higher "wage", higher avg. firm earnings, less "unemployment" [*see paper*]





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Childcare

Estimate for investments in children using childcare data in CPS (2010-present)

- Denote Childcare Expenditures (E_i) from CPS (kids $\leq$ 5yo) $\rightarrow$ positive assoc. w/ credit

$$\log(E_{i,t}) = \alpha + \beta \log(Y^P_{i,t}) + \eta \widehat{\log(C_{i,t})} + \Gamma X_{i,t} + \epsilon_{i,t}$$

$$\log(C_{i,t}) = \alpha_1 + \beta_1 \log(Y^P_{i,t}) + \eta_1 \log(AOA_{i,t}) + \Gamma_1 X_{i,t} + u_{i,t}$$

	(1)	(2)	(3)	(4)	(5)
	Dependent variable: log of parent's childcare expenditure				
Log Parent's Earnings	0.509***	0.471***	0.178	0.192	0.255*
	(0.0583)	(0.0622)	(0.178)	(0.168)	(0.143)
Log Unused Revolving Credit		0.0208**	0.182**		
		(0.0105)	(0.0897)		
Log Revolving Credit Limit				0.174**	
				(0.0836)	
Credit Score					0.930**
					(0.460)
Observations	3000	3000	3000	3000	3000
Controls	Y	Y	Y	Y	Y
Sample	ASEC	ASEC	ASEC	ASEC	ASEC
Estimation	OLS	OLS	IV	IV	IV

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Smoothing

- Credit mitigates negative effects of job loss on child earnings (OLS)
- Job loss leads to -2.1% effect on child at mean credit access, -0.5% 2SD above mean

	(1) DV: Log of child's earnings
Log Parental Earnings	0.0919***
	(0.00246)
Log Unused Revolving Credit	0.0155***
	(0.000414)
Indicator 20% Earnings Loss in Next 3 Years	-0.0350***
	(0.00462)
Log Unused Revolving Credit X Indicator 20% Earnings Loss in Next 3 Years	0.00187***
	(0.000552)
R-squared	0.236
Observations	428000
Controls	Y
Sample	Main
Marginal effect of parental earnings loss at **mean** credit	**-0.021**
Marginal effect of parental earnings loss at mean credit **plus 2SD**	**-0.005**

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Non-targeted Moments

Moment	Data Estimate	Model Estimate	Data Source
Panel I: Investment			
$1k Inc. in parent resources on children's human capital	4.10% (1.31)	6.09%	DL (2012)
Panel II: Borrowing			
$1k Inc. in unused credit on parents' borrowing	$213.00 ($80.40)	$187.79	+
Panel III: Precautionary Savings			
10% Inc. in SD of parents' income risk on consumption	-0.89% (0.42%)	-1.95%	Boar (2021)

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Credit Market Trends: Model and Data

Table: Credit Democratization Experiment Targeted Moments

Variable	Value	Target	Model	Data
		Panel I: 2000s calibration		
ψ_D	8.445	Bankruptcy rate	0.787	0.830
α_C	-0.086	Average credit limits to earnings	0.255	0.255
δ_C	-0.204	Slope, change in limits to change in earnings	-0.204	-0.204
		Panel II: 1970s calibration		
ψ_D	48.561	Bankruptcy rate	0.142	0.141
α_C	-0.001	Average credit limits to earnings	0.051	0.046*
δ_C	-0.055	Slope, change in limits to change in earnings	-0.055	-0.055**

Table: Credit Democratization Experiment Non-targeted Moments

	Model		**Data**	
	1970	2000	1970	2000
Ratio Aggregate Credit to Earnings	0.630	2.500	0.350	2.633
Average Real Interest Rate	0.051	0.083	0.082	0.109

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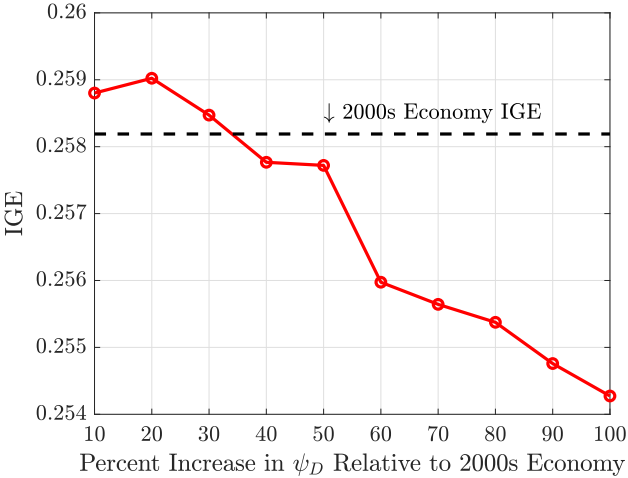
IGE by Bankruptcy Costs

Bankruptcy costs only need to ↑ by 30% to undo impact of limits



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